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SE601 p Example questions

Software Development (Virtual University of Pakistan)

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SE601p Example questions

Five Marks Questions & Answers

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These Q/A will help u cover your important topic in terms of the Q/A and clear your concept from week 1 to week 16.

1. **Explain the benefits of using an IDE in software development.**

Integrated Development Environments (IDEs) provide a suite of tools that streamline coding, debugging, and deployment processes. They offer syntax highlighting, auto-completion, error detection, and built-in debugging tools, reducing development time and improving code accuracy. Additionally, IDEs integrate with version control systems, making collaboration among developers easier and more efficient.

IDEs enhance productivity with features like code completion, debugging, and version control integration.

2. **What is the purpose of code review, and how does it improve software quality?**

Code review ensures that software is error-free, maintainable, and adheres to coding standards. It involves peer review of code to identify logic errors, security vulnerabilities, and performance issues before deployment. Regular code reviews help improve team collaboration, maintain coding consistency, and enhance software reliability by catching issues early in the development cycle.

Code reviews help detect errors, improve code maintainability, and ensure adherence to coding standards.

3. **Describe the role of build automation tools in the software development process.**

Build automation tools like Maven, Gradle, and Ant streamline the compilation, testing, and deployment of applications. They eliminate repetitive manual tasks by automating dependency management, compiling source code, running tests, and packaging applications.

These tools improve consistency, reduce human errors, and speed up the development lifecycle by ensuring smooth and efficient builds.

Build automation tools streamline compilation, dependency management, and deployment, reducing manual errors.

4. **What is the difference between a unit test and an integration test?**

A unit test verifies the functionality of individual components in isolation, ensuring that each function or method works as intended. It typically uses mock objects to simulate dependencies. In contrast, an integration test evaluates the interaction between multiple components to ensure they work together correctly. Integration testing helps identify issues related to data flow, APIs, and communication between modules.

Unit tests verify individual components, while integration tests ensure multiple components work together.

5. **How do DevOps practices contribute to software efficiency?**

DevOps integrates development and operations to streamline software delivery through automation, continuous integration (CI), and continuous deployment (CD). It reduces deployment time, enhances collaboration, and ensures faster resolution of bugs. DevOps also utilizes monitoring and logging tools to improve system reliability and scalability, leading to higher efficiency and faster software releases.

DevOps automates deployment, monitoring, and collaboration, reducing development cycles and increasing reliability.

6. **Explain the concept of continuous integration and its benefits.**

Continuous Integration (CI) is a software development practice where developers frequently merge code changes into a shared repository, triggering automated builds and tests. This approach helps identify bugs early, improves collaboration, and ensures code stability. CI minimizes integration issues by automating testing, reducing manual efforts, and enabling rapid development cycles.

Continuous integration ensures frequent code commits, automated testing, and early bug detection.

7. **What are some common debugging techniques used in software development?**

Debugging techniques include using breakpoints to halt code execution and examine variables, logging outputs to track program flow, and stepping through code to analyze logic errors. Other methods include analyzing stack traces for runtime errors, using interactive debugging tools, and employing automated debugging tools that detect memory leaks and performance bottlenecks.

Debugging involves breakpoints, logging, stack traces, and interactive debugging tools.

8. **Why is version control important in collaborative software development?**

Version control systems (VCS) like Git track changes to source code, allowing multiple developers to collaborate efficiently. VCS enables rollback to previous versions, reduces conflicts through branching and merging, and maintains a history of modifications. It enhances team productivity by facilitating code reviews, ensuring code integrity, and preventing accidental overwrites.

Version control tracks changes, enables collaboration, and provides rollback capabilities.

9. **Describe the purpose of exception handling in programming.**

Exception handling ensures that unexpected runtime errors do not crash an application. By using try-catch blocks, programs can gracefully handle errors like file not found, divide by zero, or invalid input. It improves application stability, prevents data corruption, and enhances user experience by providing meaningful error messages and recovery options.

Exception handling prevents crashes, enhances robustness, and provides meaningful error messages.

10. **What are the differences between a system exception and a business exception?**

System exceptions are related to system resources like file handling or memory allocation failures, while business exceptions are custom

exceptions specific to application logic, such as insufficient funds in a banking application.

11. **Explain the difference between static and dynamic code analysis.**

Static code analysis is performed without executing the program, checking for syntax, security vulnerabilities, and best practices. Dynamic code analysis, on the other hand, runs the program and examines its behavior during execution to detect runtime errors.

12. **How does unit testing help in maintaining software reliability?**

Unit testing ensures that individual components work correctly, allowing early bug detection and facilitating refactoring without breaking functionality.

13. **What is the importance of software documentation in a project?**

Documentation helps developers understand project structure, APIs, and workflows, ensuring easier maintenance, onboarding of new developers, and better collaboration among team members.

14. **Describe the main principles of OOP and their significance.**

The four main principles of OOP are Encapsulation (hiding internal details), Inheritance (code reuse), Polymorphism (flexibility in object behavior), and Abstraction (hiding complexity). These principles improve code modularity, reusability, and maintainability.

15. **What is object serialization, and where is it commonly used?**

Object serialization is the process of converting an object into a byte stream for storage or transmission. It is commonly used in distributed computing, saving application states, and data persistence in databases